MIL-C-45675A(ORD)
30 JULY 1962

SUPERSEDING
MIL-C-45675(ORD)
15 JULY 1960

MILITARY SPECIFICATION

COMPUTER, BALLISTICS, M13A1D

1. SCOPE

1.1 This specification covers one type of instrument known as Ballistics Computer M13A1D, used to receive the log of range data from a rangefinder calibrated for range measurement in meters, and computer superelevation for various ammunitions and transmit same to a gun sighting system in the 105mm Gun, Tank, M60.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MXLITARY

MIL–F–13926 —Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

MIL-P-14232 —Parts, Equipment, and Tools for Ordnance Material, Packaging of

MIL–F–14619 —Fire Control Parts, Equipment and Tools; Packaging of

MIL-I-45208 —Inspection Requirements, General Specification for

MTL-145607 —Inspection Equipment, Supply and Maintenance for Ordnance

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-130 —Identification Marking of U.S. Military Property

DRAWINGS

ORDNANCE CORPS

IEL7566966 —Inspection Equipment List

F8619966 —Computer, Ballistics M13A1D

LSQAP 8619966 —Supplementary Quality Assurance Provisions List

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Initial production sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regulation production.

3.2 **Materials.** Materials shall be in accordance wit h drawings, materia] specifications and general specifications forming a part of this specification.

3.3 **Construction.** The Ballistics Computer M13A1D shall be manufactured in accordance with Drawing F8619966, “Ballistics Computer, M13A1D” and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, per-

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formance and permanence to insure inherent quality and performance.

3.4 **General specification.** The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence.
(b) Protective finished surfaces.
(c) Special accuracy.
(d) Finish, painting and protective.
(e) Workmanship.

3.5 **Performance.**

3.5.1 *Output calibration.*

3.5.1.1 *Zero reference.* Input shaft rotation shall be measured from a precise 1500 meter zero reference check point previously determined by the manufacturer and indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal cover plate.

3.5.1.2 *1500 meter check point.* With the computer having a constant load of between 75and 90 ounce inches applied to the output shaft for ballistics drive, the ammunition selector shall be set on the APDS-T, M392 Cam. With the range correction knob set to zero and with the input shaft set at the 1500 meter checkpoint, actuate the superelevation hand crank to match the outer index with the inner index within plus or minus half a line width, by approaching this range first from one direction then from the opposite direction, without over travel. Note the position of the output shaft for ballistics drive and split the backlash error atthis output shaft. This is then the check point for computer output. The position of the superelevation counter shall be established from this check point and shall conform to the requirements of Table I.

TABLE I *Position of Superelevation Counter.*

Output shaft position	177.84° ccw	1500 meter check point 0°	167.76° cw	1R 23.76° cw
Tolerance	±8°	±3°	±8°	±8°
Superelevation Counter Mils	1.20	3.67	6.00	9.00

3.5.1.3 *Outer index and inner index.* Using the APDS-T, M392 Cam only, position the input shaft to correspond with each of the ranges as indicatedin table II and rotate the hand crank until the outer index is in coincidence with the inner index, at which point the rotation of this output shaft shall correspond to the values in tables II and III. This procedure shall be repeated to check each cam at 1500 meters only. All checks shall be made by approaching the index from both clockwise and counterclockwise direction.

3.5.1.4 *Backlash.* Backlash of the output shaft for the superelevator shall not exceed a total of one degree at any point of rotation of the shaft.

3.5.1.5 *Outer index and range scale.* Using the APDST, M392 Cam only, rotate the hand crank until the outer index is in coincidence with the selected range on the computer range scale for each of the ranges as indicated in table II, at which point the rotation of the output shaft for ballistics drive shall correspond to the values in tables II and III. All checks shall be made by approaching the index from both clockwise and counterclockwise direction.

3.5.1.6 *Servo follow-up sensitivity.* Withan output shaft load of belween 75 and 90 ounce inches applied to the output shaft for the ballistics drive, the ammunition selector shall be set onthe APDS-T, M392 Cam, the power on at 18 volts plus or minus 0.5 volts, and the automatic follow-up cycle completed at ranges between 3800 and 4400 meters, rotate the input shaft slowly so that it is possible to count the number of times the contact closes in order to advance the output shaft to the ballistics drive by approximately 72 degrees. Calculate the average change in superelevation per one make and break of the contacts. Repeat in the opposite direction. The average allowable superelevation increment for each range shall not exceed 10 degrees 45 minutes and the maximum superelevation increment shall not exceed 21 degrees 30 minutes as read at the output shaft to the ballistics drive.

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3.5.2 *Power operation.* With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft for the ballistics drive, the rotation of this output shaft shall correspond to the values indicated in tables II and III for the ammunition and ranges specified therein. This requirement shall be met by allowing the outer index to approach the inner index from both increasing and decreasing range with power on continuously at 28 volts plus or minus 0.5 volts. The computer shall also meet the accuracy requirement of tables II and IIIfor the following ammunition and at ranges indicated when checked at 18 volts plus or minus 0.5 volts.

Ammunition and Rangcjor18 Volt Check

Ammunition	*Ranges*
APDS-T, M392	500, 1000, 2000 and3600 meters
HEP-T, M393	500, 1000, 2000 and 3600 meters
HEAT-T, M456, (T384E4)	500, 1000, 2000 and 3600 meters

All checks shall be made by approaching index from both clockwise and counterclockwise direction.

TABLE II. *Computer Accuracy as Determined by Position of Rotation of the Output Shaft Based on 1500 Meter Reference Ranoe for APDS-T, M392 Cam (Drawing No. 8619776).*

Range (meters)	500	1000	1200	1500	2000	3000	3300	3600	4400
Input Shaft Position (±1°)	11R 68.54° cw	4R 46.80° cw	2R 98.25° cw	0° Check Point	2R 334.91° ccw	7R 21.71° ccw	8R 11.23° ccw	8R 330.30° ccw	10R 346.14° ccw
APDS-T, M392	0R 182.16° ccw *1.14	0R 93.60° ccw *2.37	0R 56.88° ccw *2.88	0° *3.67	0R 95.76° cw *5.00	0R 300.24° cw *7.84	1R 5.76° cw *8.75	1R 74.88° cw *9.71	1R 282.24° cw *12.59
HEP-T, M393	0R 216.72° cw *6.68	1R 324.72° cw *13.18	2R 182.16° cw *16.20	3R 187.92° cw *21.28	5R 197.28° cw *31.41	11R 44.64° cw *59.29	13R 94.32° cw *69.98	15R 227.52° cw *81.83	....
HEAT-T, M456 (T384E4)	0R 122.40° ccw *1.97	0R 48.24° cw *4.34	0R 126.00° cw *5.42	0R 255.60° cw *7.22	1R 151.20° cw *10.77	3R 183.60° cw *21.22	4R 144.72° cw *25.68	5R 172.08° cw *31.06	9R 232.56° cw *51.90

(*) Corresponding nominal superelevation mil value (see 6.4.3).

3.5.2.1 *Output shaft tolerances.* When the output shaft rotation of the computer, as read in degrees, is equivalent to the superelevation value in mils as shown in table III, column A, the output shaft tolerance shall be within the corresponding value as shown in column B.

3.5.3 *Stability.* Using 28 volts plus or minus 0.5 volts, with a load of between 14 and 18 ounce inches applied to the output shaft to ballistics drive, ballistics correction set to introduce minus 15 percent within 1 percent, the rotation of the input shaft between minimum range and 1500 meters shall not

TABLE III. *Output Shaft Tolerances.*

A Output shaft rotation	B Degrees	C Mils*
0 up to but less than 6	± 7.20	±0.10
6 up to but less than 11	±10.80	±0.15
11 up to but less than 26	±14.40	±0.20
26 up to but less than 31	±18.00	±0.25
31 up to but less than 36	±25.20	±0.35
36 up to but less than 41	±28.80	±0.40
41 up to but less than 61	±36.00	±0.50
61 up to but less than 86	±43.20	±0.60
86 up to but less than 98.5	±50.40	±0.70

(*) The data contained in column C is informational only. (See 6.4.3)

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cause the computer servo to display any self-sustaining or oscillatory operation. This check is required for all cams. A maximum of eight changes in direction of the servo motor, as a settling out action, shall be considered as satisfactory operation. These eight changes may be counted as clicks.

3.5.4 *Component operation.*

3.5.4.1 Operational values shall not exceed those given intable IV at the temperatures indicated.

Table IV. *Operational Values*

Control	Operation	Room temp. to 150° F.	Minus 40° F.
Input shaft	Rotate	16 oz. in.	45 oz. in.
Range correction knob	Pull Out	20 lbs.	24 lbs.
	Rotate	34 oz. in.	45 oz. in.
Ammunition selector	Rotate	60 lb. in.	80 lb. in.
	Translate	10 lbs.	20 lbs.
Handcrank	Hold engaged	12 lbs.	20 lbs.
	Rotate engaged	6 lb. in.	20 lb. in.
Reset button	See par. 3.6.3	15 lbs.	40 lbs.
Iris dimmer	Rotate	Mechanically operable	
Interrupter switches	See par. 3.6.3	Mechanically and electrically operable	

3.5.4.2 *Time of response.* At standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) with the output shaft load as specified in 3.5.2 applied, the inner and outer indexes positioned to the values in table V, and the ammunition selector set on the APDS-T, M392 Cam, the outer index shall stabilize on the inner index within the elapsed time specified after power has been applied.

3.5.4.3 *Circuit breaker.*

3.5.4.3.1 *Breaker opens.* With the motor at standard ambient temperature and with 28 volts plus or minus 0.5 volts applied, the two indexes separated by at least 200 meters, the output shaft for the ballistics drive locked

TABLE V. *Time of Response.*

Outer index position (meters)	Inner index position (meters)	Voltage (plus or minus 0.5V)	Total elapsed time (seconds)
4400	500	28	4
500	4400	28	4
500	4400	18	8
4400	500	18	8

(immovable), the circuit breaker shall open in 90 seconds or less.

3.5.4.3.2 *Breaker closed.* At standard ambient temperature with the load as specified in 3.5.2 applied to the output shaft for the ballistics drive, the computer operating at 28 volts plus or minus 0.5 volts, using the APDS-T, M392 Cam, the input shaft shall be oscillated for a period of forty-five seconds over a spread of 1000 meters, plus or minus 100 meters between ranges of 1200 to 2800 meters, at a rate of 4 to 4.5 revolutions per second, during which time the circuit breaker shall not open.

3.5.4.4 *Hand crank and input shaft travel.*

3.5.4.4.1 The rotation of the superelevation hand crank within the limits of its stops shall cover the excursion from minus 1.5 mils plus or minus 0.1 roils to plus 98.5 mils plus or minus 0.5 mil as indicated on the superelevation counter when the ballistics correction knob is set at zero.

3.5.4.4.2 The range input shaft shall rotate through an excursion of 24.0 revolutions plus 0.1 revolution, and shall cover the range from 480 to 4575 meters within this excursion, as measured from the 1500 meter check point of the input shaft (see 6.5).

3.5.5 *Superelevation cams.* The cams shall be placed in the ballistics computer in the following sequence starting with the APDS-T, M392, followed by the HEP-T, M393, followed by the HEAT-T, M456 (T384E4), followed by three spacers. The computer shall be shipped with the same cams with which it was qualified.

3.6 **Electrical check.**

3.6.1 *Insulation breakdown.* Insulation

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breakdown shall be in accordance with Specification MIL-G-13926 except that breakdown voltage of 500 volts root mean square (rms) shall besubstituted for 750 volts rms.

3.6.2 *Insulation resistance.* Insulation resistance shall be in accordance with Specification MIL-F-13926 except that the resistance value shall not be less than 2 megohms and a direct current (dc) potential of 500 volts shall be used in place of 800 volts. All lamps shall be removed from sockets prior to tests.

3.6.3 *Continuity.* The electrical circuits shall be checked for continuity and shall conform to the wiring diagram. When the hand crank is engaged, the interrupter switch shall open the motor circuit and energize the pilot light. Pressing of the reset button shall cause the motor circuit to be closed and the pilot light to be extinguished. Actuation of the ammo selector handle shall cause the ammunition selector interrupter switch to open the motor circuit. Resetting of the handle shall cause the switch to complete the circuit.

3.6.4 *Illumination.* The internal lighting device shall provide adequate illumination so that the inner and outer indexes of the range dial and the ammunition index shall be clearly defined and readable when observed in a darkened enclosure.

3.7 **Environmental.**

3.7.1 *Storage temperatures.* The computer shall show no evidence of physical failure and shall meet the requirements of 3.7.2.1 at standard ambient temperature after having been exposed and thermally stabilized at ambient temperatures of minus 80 degrees and plus 160 degrees Fahrenheit.

3.7.2 *Extreme temperature operability.* when exposed and thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit, the computer shall meet the following requirements:

(a) Applicable requirements of 3.5.4.1 at temperatures indicated in Table IV.

(b) At minus 40 degrees Fahrenheit, the outer index shall follow the inner index within one-half a graduation width over the full excursion of the range scale in a total elapsed time of 8 seconds for each direction when checked with 18 volts, plus or minus 0.5 volts.

(c) At plus 150 degrees Fahrenheit using 28 volts, plus or minus 0.5 volts, the ammunition selector set on the APDS-T, M392 cam, with a load of between 14 and 18 ounce inches applied to the output shaft for the ballistics drive and with the indexes separated by 3600 meters, the outer index shall stabilize on the inner index and shall meet the applicable requirements as specified in 3.5.3 when the switch is thrown to the “ON” position. Immediately upon completion of the environmental requirements as specified in 3.7, the interior of the computer shall be dried by clean, warm air of between 100 degrees to 140 degrees Fahrenheit.

3.7.2.1 *Post environmental check.* Each computer which has been subjected to the environmental conditions as specified in 3.7.1 and 3.7.2 shall meet the following requirements after thermal stabilization has been re-established at standard ambient temperature.

(a) The requirements of table I and 3.5.3.

(b) When checked under power operation at 28 volts, plus or minus 0.5 volts, with 75 to 90 ounce inches load applied, the output shaft for the ballistics drive shall indicate the rotational values of table II within the tolerances of table III for the following ammunition and ranges:

Ammunition	*Ranges*
APDS-T, M392	500, 1000, 2000 and 3600 meters
HEP-T, M393	500, 1000, 2000 and 3600 meters
HEAT-T, M456 (T384E4)	500, 1000, 2000 and 3600 meters

3.7.3 *Vibration.* With the ammunition selector set on the APDS-T, M392 cam, the

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outer index set at 3000 meters, the computer (less any shock mounts and brackets, universal assembly B8200862 and output shaft B8200801) shall be vibrated at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a durationof5 minutes, p1us or minus15 seconds.

3.7.3.1 *Postvibration.* Subsequent to vibration, the computer shall not be damaged or show evidence of broken, loose or missing parts and it shall meet the requirements of this specification.

3.7.4 *Sealing.* Thecomputer shall meetthe requirements of the rain test of Specification MIL-F-13926.

3.8 **Identification marking.** Identification marking, unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibilities.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conforming to prescribed requirements.

4.2 **General requirements.** Inspection requirements and procedures shall be in accordance with Specification MIL-I-45208.

4.3 **initial production sample.** Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 *Productionsample failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. Where corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 **Lot formation.** Inspection lotsizes, lot formation and sampling for inspection shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 **Material.** Control ofmaterial shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 **Inspection of components.** Inspection requirements for component parts and sub assemblies are contained herein and on Supplementary Quality Assurance Provision (SQAP) listed on SQAP List (LSQAP) 8619966.

4.6.1 *Contents of SQAP.* The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual nonconforming items.

4.6.2 *Contractor's inspection for all components not covered by SQAP.* Inspection of these components shall be in accordance with inspection procedures of the written inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional

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characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional performance of the end item.

4.7 **Inspection equipment.** The inspection equipment shall conform to that listed in IEL 7566966 except as provided for under the alternate inspection procedures and inspection equipment requirements of Specification MIL-I-45208. The provisions of Specification MIL-I-45607 shall apply to the test equipment provided by the Government or providedbythe contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

4.7.1 *Accuracy requirements.* Inspection equipment used in the performance of this specification shall be capable of measuring withinan accuracy of 10 percent of thespecified product requirement tolerance. In cases where this requirement is seemingly impractical, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8 **Acceptance inspection.**

4.8.1 *Examination.* Classification of defects in table VI shall constitute the contractor's minimum examination to be performed for acceptance or rejection by lot.

TABLE VI, *Classification of Defects.*

Use Inspection Level II in table IIIA with Sampling Plan Table IVA of Standard MIL-STD-105

Critical: None defined.

Major:	AQL 1.5 percent defective	Req't. par.	Q.A. par.	Test method
101	Physical failure, post vibration	3.7.3.1	4.9.4.1	Visual-Tactile
102	Output shaft	3.5.1.2	4.9.5.1	Special testing equipment (STE)
103	Outer index and inner index	3.5.1.3	4.9.5.2	STE
104	Backlash	3.5.1.4	4.9.5.3	STE
105	Outer index and range scale	3.5.1.5	4.9.5.4	STE
106	Servo follow-up sensitivity	3.5.1.6	4.9.5.5	STE
107	Power operation and output shaft tolerance	3.5.2	4.9.5.6	STE
108	Stability	3.5.3	4.9.5.7	STE
109	Time of response	3.5.4.2	4.9.5.8	STE
110	Circuit breaker closed	3.5.4.3.2	4.9.5.9.1	STE
111	Circuit breaker opens	3.5.4.3.1	4.9.5.9.2	STE
112	Hand crank travel	3.5.4.4.1	4.9.5.10.1	STE
113	Input shaft travel	3.5.4.4.2	4.9.5.10.2	STE
114	Superelevation cams	3.5.5	4.9.5.11	Visual
115	Operating values	3.5.4.1	4.9.6	STE
116	Insulation breakdown	3.6.1	4.9.7.1	Standard Measuring Equipment (SME)
117	Insulation resistance	3.6.2	4.9.7.2	SME
118	Continuity	3.6.3	4.9.7.3	SME
119	Illumination	3.6.4	4.9.8	Visual
120	Identification marking	3.8	4.9.10	Visual
Minor:	None defined.			

Note. The above examinations shall be conducted at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) in accordance with the quality assurance paragraph (Q.A. par.) listed.

4.8.2 *Tests.* The tests listed in table VII shall constitute the contractor's minimum testing to be performed prior to acceptance or rejection by lots.

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TABLE VII. *Inspection Testing*

Use Inspection Level L6 in table IIIB with Sampling Plan table IVA of Standard MIL-STD-105

Critical: None defined.

Major:	AQL 1.5 percent defective	Req't. par.	Q.A. par.	Test method
101	Stability (–40° F. and +150° F.)	3.7.2 (c)	4.9.1.2	STE
102	Outer index and inner index (–40° F.)	3.7.2 (b)	4.9.1.3	STE
103	Outer index and inner index (+150° F.)	3.7.2 (b)	4.9.1.4	STE
104	Operating torques (–40° F. and +150° F.)	3.7.2 (a)	4.9.1.5	STE
105	Physical failure, post environmental	3.7.1	4.9.1.6	Visual
106	Output shaft, post environmental	3.7.2.1 3.5.3	4.9.2.1	STE
107	Stability, post environmental	3.7.2.1 3.5.3	4.9.2.2	STE
108	Power operation, post environmental	3.7.2.1	4.9.2.3	STE
109	Sealing (water resistance)	3.7.4	4.9.3	Visual
Minor:	None defined.			

Note. The tests for characteristics 101, 102, 103 and 104 shall be conducted at the temperatures indicated. The tests for characteristics 105, 106, 107, 108 and 109 shall be conducted at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) subsequent to the temperature cycling test. All tests shall be conducted in accordance with the quality assurance paragraph (Q.A. par.) listed.

4.9 **Test methods and procedures.**

4.9.1 *Environmental test.* Samples for environmental testing shall be selected as specified. The contractor's Government approved refrigeration, heating and rain test chambers shall be used to perform the environmental test.

4.9.1.1 The operational temperature test may be conducted concurrent with the storage temperature test. Caution must be exercised so as not to expose the computer to thermal shock during the test. Temperature changes shall be gradual.

4.9.1.2 *Stability.* The computer shall be subjected to this test when thermal equilibrium of the computer is established at minus 40 degrees and plus 150 degrees Fahrenheit, respectively. With the ammunition selector set on the cam as specified in 3.5.1.2, this test shall be conducted as specified in 4.9.5.7. The computer shall meet the requirements as specified in 3.5.3 and 3.7.2.

4.9.1.3 *Outer index and inner index.* When thermal equilibrium of the computer is reached at minus 40 degrees Fahrenheit, the computer outer index shall meet the specified requirements when the inner index is translated over the full excursion of the range stile within the time limit specified. This translation of the inner index shall be accomplished by a continuous rotation of the computer input shaft from the clockwise stop to the counterclockwise stop. Repeat test by rotating the input. shaft in the opposite direction. This test shall reconducted with the specified voltage applied to the computer and the circuit breaker switch in the "ON''position.

4.9.1.4 *Outer index and inner index.* when thermal equilibrium of the computer is reached at plus 150 degrees Fahrenheit, set the ammunition selector on the cam specified, the range correction knob to zero and apply the specified torque load to the computer output shaft. Position thecomputer inner index at 500 meters and the outer index at 4400 meters on the range scale. When the circuit breaker switch is thrown to the "ON" position, the outer index shall "settle out" (ceases oscillatory operation) coincident to the inner index within eight (8) changes (clicks) in the direction of the computer

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servo. The computer shall again meet this requirement when the test is repeated with the outer index positioned at 500 meters and the inner index positioned at4400 meterson the range scale.

4.9.1.5 *Operating torques and forces.* When thermal equilibrium of the computer is reached at minus 40 degrees and plus 150 degrees Fahrenheit, respectively, the controls listed in table IV shall not exceed the applicable torque and force values specified therein for the operation of those controls. This test shall be conducted as specified in 4.9.6.

4.9.1.6 *Physical failure, environmental.* Each computer subjected to the storage temperatures shall be examined for physical failure of castings, mechanical parts, electrical parts and glass breakage. This examination shall be conducted at standard ambient temperature after exposure to each temperature extreme.

4.9.1.7 *Drying.* Immediately following the environmental test, the computers subjected. to that test shall be air dried as specified.

4.9.2 *Post environmental check.* Subsequent to the environmental check, the computer subjected to that test shall be set up as specified in 4.9.5.

4.9.2.1 *Output shaft, post environmental.* This test shall be conducted as specified in 4.9.5 and 4.9.5.1 and shall meet the requirements of table I.

4.9.2.2 *Stability, post environmental.* This test shall be conducted as specified in 4.9.5 and 4.9.5.7 and shall meet the specified requirements.

4.9.2.3 *Power operation, post environmental.* This test shall be conducted as specified in 4.9.5, 4.9.5.6 and 4.9.5.6.1 for the camsand ranges specified. The results ofthis test shall meet the applicable requirements of tables IIand III.

4.9.3 *Sealing (water resistance).* This test shall be conducted in accordance with procedure I of Specification MIL-F-13926. The computer shall show no evidence of water penetration as a result ofthistest. The receptacle 8200970 shall be covered with waterproof tape during the test.

4.9.3.1 The computers that are opened for environmental drying and examination for water penetration shall be resealed to meet the requirements of the drawings prior to submission for final acceptance.

4.9.4 *Vibration.* Eachcomputerinthe sample lot shall be vibrated as specified in 3.7.3. This operation shall be performed on fixture F7650085 using adapter F7659608.

4.9.4.1 *Physical failure, post vibration.* As a result of vibration, the computer shall show no evidence of physical failure of mechanical parts or glass breakage.

4.9.5 The following tests shall reconducted with the computer positioned on test fixture F7659607 and in accordance with the operating instructions specified on drawing F7659607. The computer input shaft and the fixture input mechanism shall be oriented to the 1500 meter check point as outlined in the set-up instructions on drawing F7659607.

4.9.5.1 *1500 meter check point.* Setthe ammunition selector on the cam specified and the range correction knob on zero. When the mil value specified in table I is indicated on the computer superelevation mil counter, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified in table I.

4.9.5.2 *Outer index and inner index.* Set the ammunition selector on the cam specified and the range correction knob on zero. When the rotational position of the computer input shaft corresponds to the value for each of the ranges indicated in table II, and the computerouter and inner indexes are coincident, the rotational position of the ballistics drive output shaft shall not exceed the tolerance specified for the applicable values in tables IIand III Each cam shall meet the requirements of tables II and III for the specified range.

4.9.5.3 *Backlash.* Backlash of the computer superelevation output shaft shall not exceed

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the tolerance when measured at four (4) points of shaft rotation. The four (4) points of measure shall be approximately 90 degrees apart.

4.9.5.4 *Outer index and range scale.* Setthe ammunition selector on the cam specified and the range correction knob on zero. When the computer outer index is coincident to the selector range mark on the computer range scale, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified for the applicable values in tables II and III.

4.9.5.5 *Servo follow-up sensitivity.* Set the ammunition selector on the cam specified, the range correction knob on zero and the inner and outer index to a range between 3800 meters and 4400 meters. With the computer in power operation, the average and maximum recorded increment change (closing of computer servo contracts) resulting from atotal rotation of theoutputshaft72 degrees clockwise and 72 degrees counterclockwise shall not exceed the tolerance specified. Superelevation increment changes shall be measured at the computer output shaft for the ballistics drive.

4.9.5.6 *Power operation.* This test shall be conducted as specified in drawing F7659607 with the range correction knob set to zero.

4.9.5.6.1 With the input voltage set at 28 volts, plus or minus 0.5 volts, and the circuit breaker switch in the "ON'' position,theaccuracy of each cam shall be tested at the ranges listed in table II for computer output shaft rotation. When the input shaft is rotated to a value indicated in table II for a given range, the rotational position of the output shaft shall correspond to the value for that range in table II within the tolerance specified in table III.

4.9.5.6.2 With the input voltage set at 18 volts, plus or minus 0.5 volts, the computer shall meet the accuracy requirements of tables II and III for the cams and ranges.

4.9.5.7 *Stability.* Prepare the computer for this test in accordance with Drawing F7659607, set the range correction knob to minus 15 percent within 1 percent and adjust the input voltage to 28 volts, plus or minus 0.5 volts. Rotate the computer inner index 1500 meters; throw the circuit breaker switch to "ON" position to energize the servo motor. The outer index shall "settle out" (cease oscillatory operations) coincident to the inner index within 8 changes (clicks) in direction of the computer servo. The inner index shall then be rotated to 4 different ranges between 1500 meters and minimum range, and the outer index again shall "settle out" coincident to the inner index within 8 changes in direction of the servo. Set the range correction knob to plus 15 percent within 1 percent and repeat the aforementioned test. All cams installed in or accompanying the computer shall be subjected to this test.

4.9.5.8 *Time of response.* Set the ammunition selector on the sam specified and the range correction knob on zero. When the circuit breaker switch is thrown to the "ON" position and the servo motor is energized, the computer outer index shall stabilize on the inner index within the time cycle specified in table V.

4.9.5.9 *Circuit breaker.*

4.9.5.9.1 *Circuit breaker, closed.* Set the ammunition selector on the cam specified, the range correction knob on zero and the inner and outer index coincident on 2000 meters of range. The computer input shaft shall be oscillated as specified with the circuit breaker switch in the "ON" position and the servo motor energized. The circuit breaker shall not open (switch move to "OFF" position) as a result of this oscillation.

4.9.5.9.2 *Circuit breaker, opens.* Set the ammunition selector and range correction knob as specified in 4.9.5.9.1. The inner and outer index shall be separated by the specified distance. When the circuit breaker switch is thrown to the "ON" position and the servo motor is energized, the circuit breaker shall open (switch move to "OFF" position) within the specified time cycle.

4.9.5.10 *Hand crank and input shaft travel.*

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4.9.5.10.1 The test for hand crank travel shall be conducted in accordance with drawing F7659607 and shall meet the specified requirements.

4.9.5.10.2 The test for input shaft travel shall be conducted in accordance with drawing F7659607 and shall meet the specified requirements.

4.9.5.11 *Superelevation cams.* The inspection sequence and arrangement of cams in the computer shall be as specified. The process of replacing cams in the computer shall be as specified in drawing F7659607.

4.9.6 *Operating torques and force.* The controls listed in table IV shall not exceed the applicable torque and force values specified therein for the operation of those controls while at an ambient temperature of plus 60 degrees to 90 degrees Fahrenheit. The equipment to be utilized for this test is listed on drawing D8570433. The reset button shall be inspected by use of standard measuring equipment (SME). This standard measuring equipment shall be capable of measuring the values within one-quarter (¼) pound.

4.9.7 *Electrical check.* The following tests shall be conducted at any time subsequent to vibration. Before testing, remove lamp bulbs and disconnect capacitor from computer circuit.

4.9.7.1 *Insulation breakdown.* This test shall be conducted in accordance with Specification MIL-F-13926 and shall meet the specified requirements.

4.9.7.2 *Insulation resistance.* This test shall be conducted in accordance with Specification MIL-F-13926 and shall meet the specified requirements.

4.9.7.3 *Continuity.* A continuity test of the computer circuits shall be made using standard measuring equipment to determine conformance with the wiring diagram. Actuation of the hand crank, reset button and ammunition selector handle shall cause the computer to function.

4.9.8 *Illumination.* The illuminated inner and outer indexes of the range dial and ammunition index of the computer shall meet the requirements when observed in a darkened enclosure with the unaided eye.

4.9.9 *Final acceptance.* Final acceptance of the lot of computers shall be withheld by the Government inspector until the sample lots of computers, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

4.9.10 *Identification marking.* Marking shall be visually inspected for conformance to Standard MIL-STD-130.

4.10 **Quality assurance provisions for packaging, packing and marking.** Quality assurance provisions for packaging, packing and marking of the Computer, Ballistics, M13A1D shall be in accordance with 5.1 and Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 **Packaging, packing and marking.** The computer shall be packaged, packed and marked in accordance with Specification MIL-F-14619/P8619966. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 **Intended use.** The Ballistics Computer M13A1D is an electro-mechanical component of the primary fire control system for the 104mm Gun, Tank M60. The ballistics computer manually or electrically determines and relays to the sighting system the amount of gun superelevation required for specific target range and ammunition type.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model, including date of delivery (see 3.1).

6.3 **Design information.** Information regarding specific design not covered or ref-

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erenced by this specification may be obtained from theprocuring activity upon specific request of the contractor.

6.4 **Definitions.**

6.4.1 *Special testing equipment (STE).* Special inspection fixtures, instruments and modified measuring equipment and special purpose commercially available equipment.

6.4.2 *Standard measuring equipment (SME).* Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items) and which are normally used by an inspector to perform dimensional inspection of items under procurement, that is, micrometers, rigid gages, gage blocks, depth gages, scale protractors, dial indicators, voltmeters, ohm meters, etc.

6.4.3 *Superelevation values in mils.* The mil values in table II represent the corresponding nominal superelevation values in roils for the calculated rotational values in the table. The actual superelevation value in mils that will correspond to computer output rotation will be obtained by modifying the nominal value by the relationship that each 7.20 degrees of rotation of the output shaft above or below the nominal value is equal to 0.10 mil.

6.5 **Input shaft excursion.**

From 1500 meter zero check point

11R	218° CW = 480 meters.
10R	346° CCW = 4400 meters.
11R	129° CCW = 4575 meters.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

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